

[Menu](#)

Overview

▼ Relevant source files

`admin/utilities/downloadInteractionLog.php``configIni.php``hclient/core/layout.js``hclient/framecontent/initPage.php``index.php``layout_default.js``movetoparent/heuristConfigIni.php``redirects/resolver.php``viewers/record/viewRecord.php`

The Heurist Network system is an academic knowledge management platform designed to store, manage, analyze, and publish research data. This document provides a high-level overview of Heurist's architecture, core components, and how they interact to deliver its functionality.

Heurist allows researchers to create customized database structures without programming knowledge, import data from various formats, perform complex searches, analyze relationships between records, and publish data as interactive websites, visualizations, or data exports.

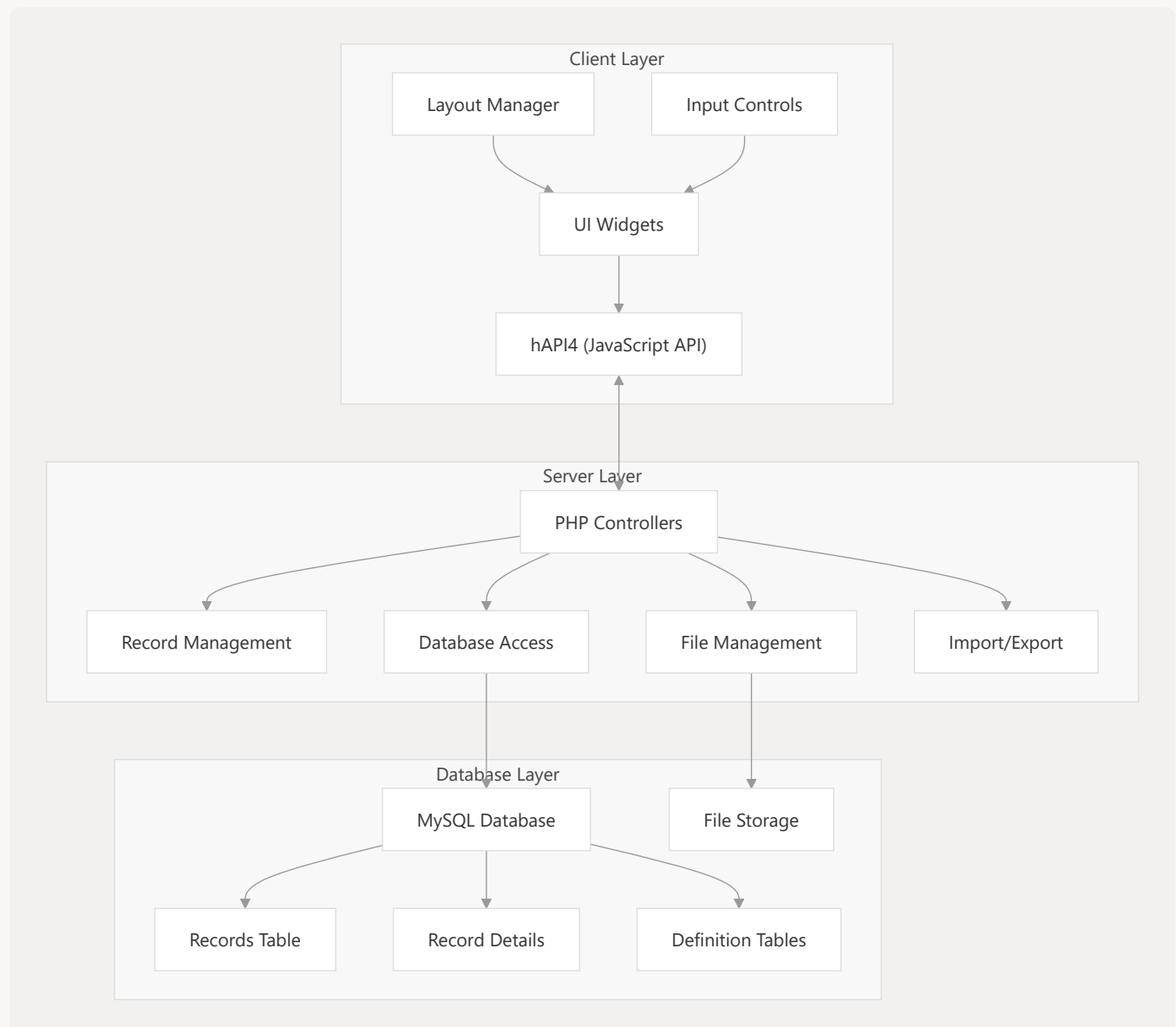
System Architecture

Heurist follows a client-server architecture with a modular design that separates core functionality from specialized features. The system is built around a central database with flexible record types and fields that can be customized for specific research domains.

High-Level Architecture

Sources: `configIni.php` `index.php` `layout_default.js`

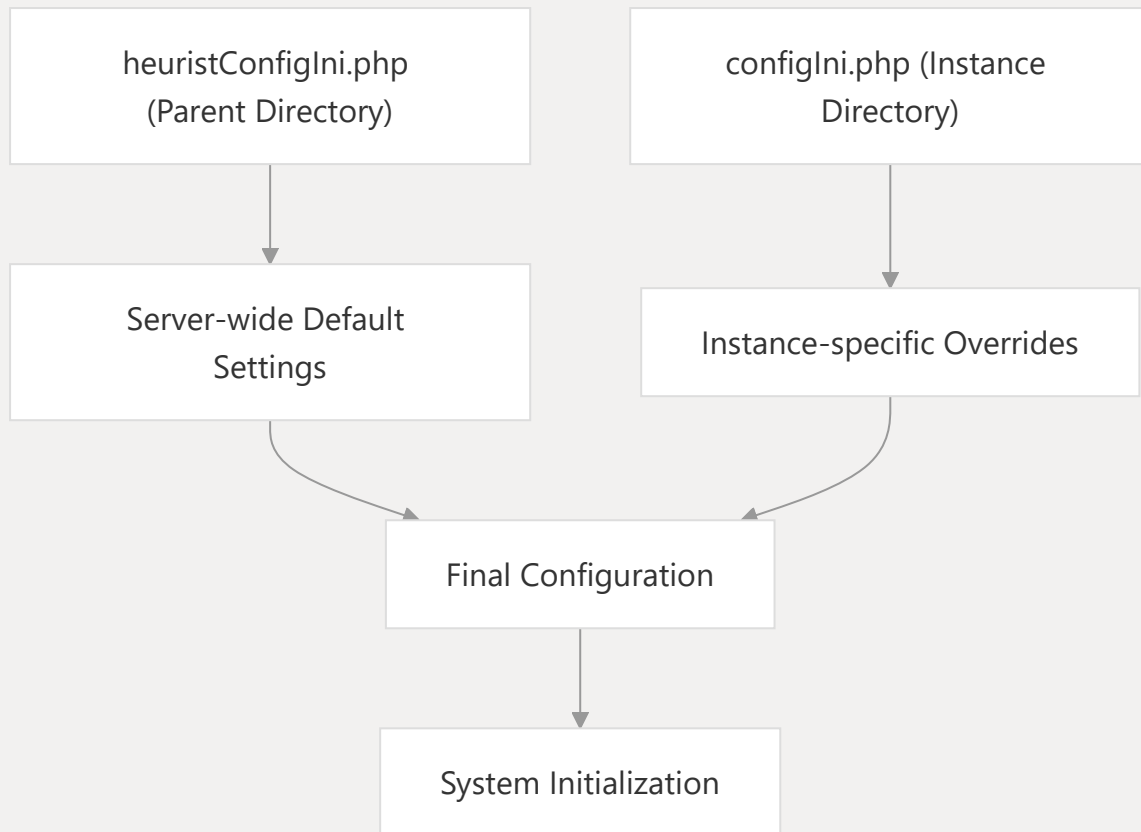
Client-Server Architecture

Sources: `index.php` 21-27 `hclient/framecontent/initPage.php` 219-235
`layout_default.js` 99-115

Configuration System

Heurist uses a hierarchical configuration system that allows for server-wide defaults with instance-specific overrides.

Configuration Hierarchy



Sources: `configIni.php` 29-42 `configIni.php` 206-214
`movetoparent/heuristConfigIni.php` 24-35

Key configuration settings include:

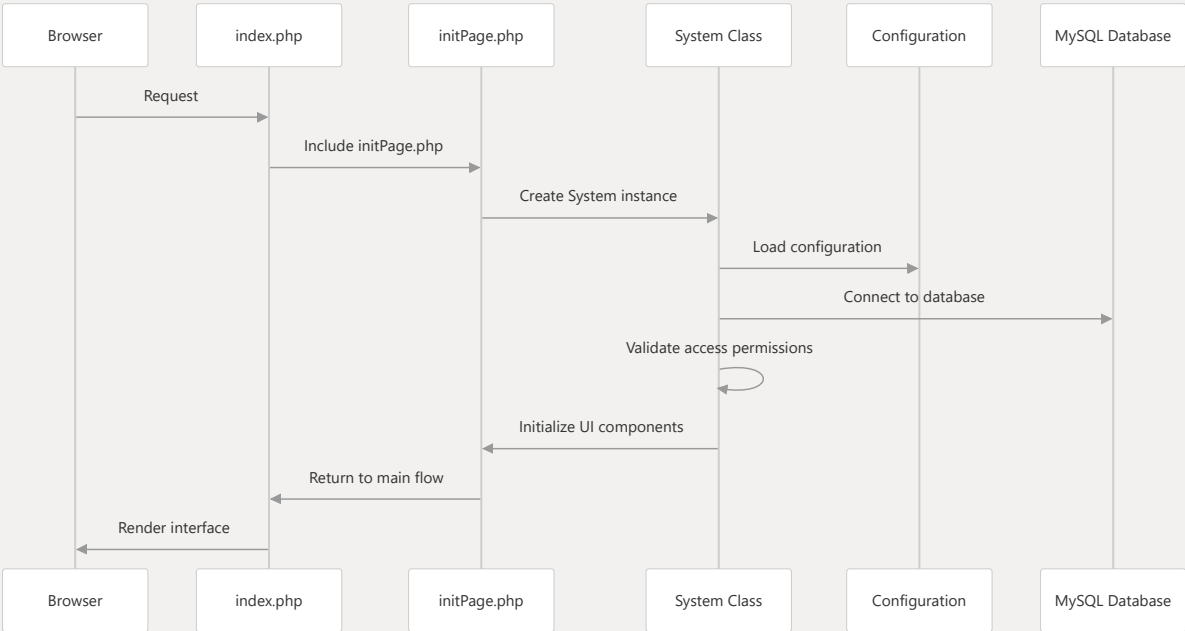
- Database connection parameters
- File storage paths
- Email settings
- Security settings
- UI customization options
- Integration with external services

The configuration system uses a parent-child relationship between configuration files. The `heuristConfigIni.php` in the parent directory provides default settings, while `configIni.php` in each Heurist instance can override these settings as needed.

Core Components

Initialization Process

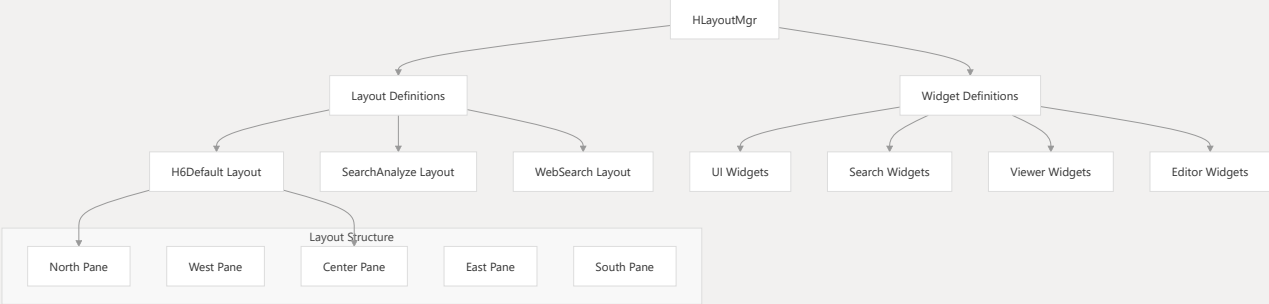
The system initialization follows a specific sequence to ensure all components are properly loaded and configured.



Sources: `index.php` 25-26 `hclient/framecontent/initPage.php` 44-46
`hclient/framecontent/initPage.php` 297-305

Layout Management

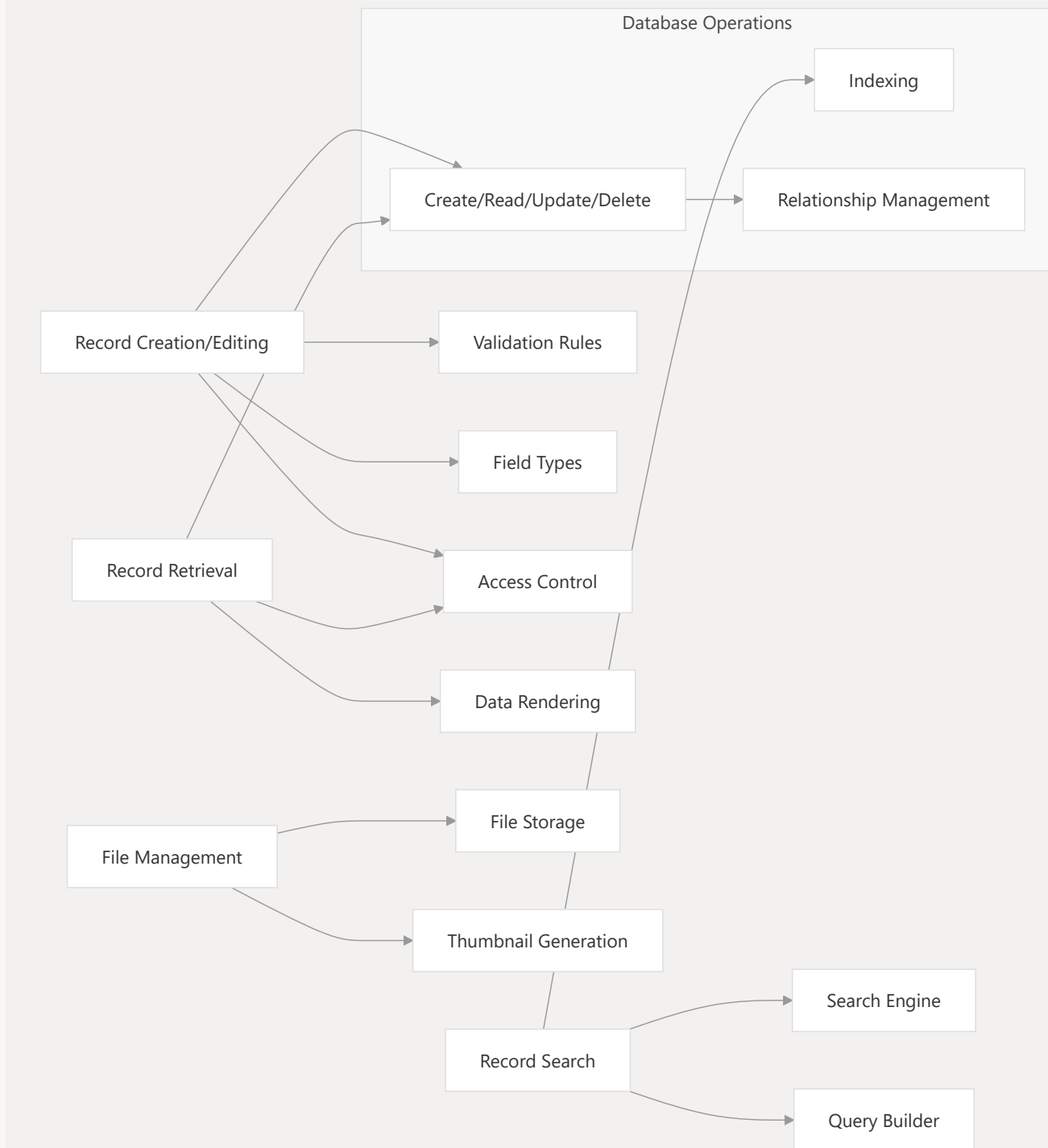
Heurist uses a flexible layout system that supports various UI configurations. The layout system is defined in `layout_default.js` and managed by the `HLayout` class in `layout.js`.



Sources: `layout_default.js` 40-79 `layout_default.js` 99-135 `hclient/core/layout.js` 29-45

Data Flow

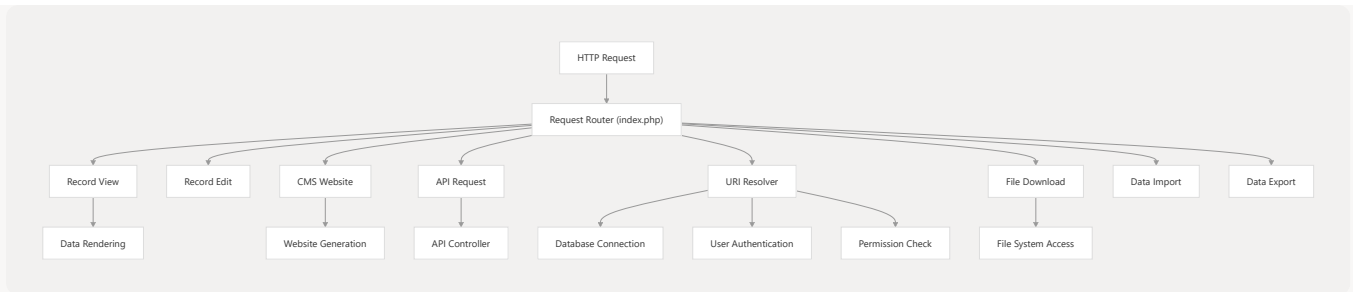
Record Management



Sources: `index.php` 43-55 `index.php` 65-103 `viewers/record/viewRecord.php` 60-83

Request Handling

Heurist processes requests through a central request handler that determines the appropriate action based on URL parameters.

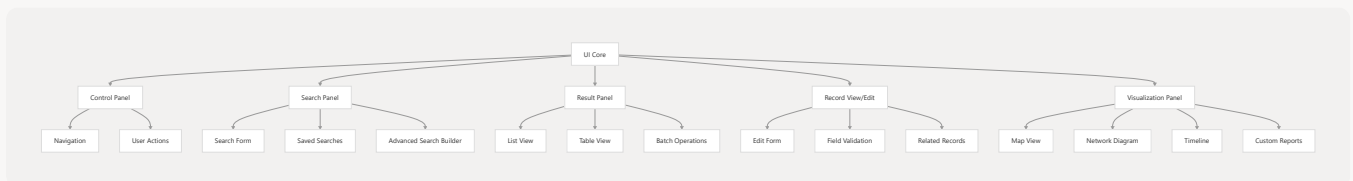


Sources: `index.php` 44-151 `redirects/resolver.php` 101-206 `redirects/resolver.php` 400-456

User Interface Components

Heurist's UI is built on a widget-based system with configurable layouts. The primary interface components include:

1. **Control Panel** - Main navigation and database management tools
2. **Search Interface** - For querying and filtering records
3. **Result List** - For displaying search results
4. **Record View/Edit** - For viewing and editing record details
5. **Visualization Tools** - Maps, networks, and other visualizations



Sources: `layout_default.js` 119-127 `layout_default.js` 210-247 `index.php` 249-293

Data Import and Export

Heurist provides robust data import and export capabilities to facilitate data exchange with other systems.

Import Formats

- CSV/TSV
- XML/JSON
- HML (Heurist Markup Language)
- Zotero
- BibTeX
- RDF

Export Formats

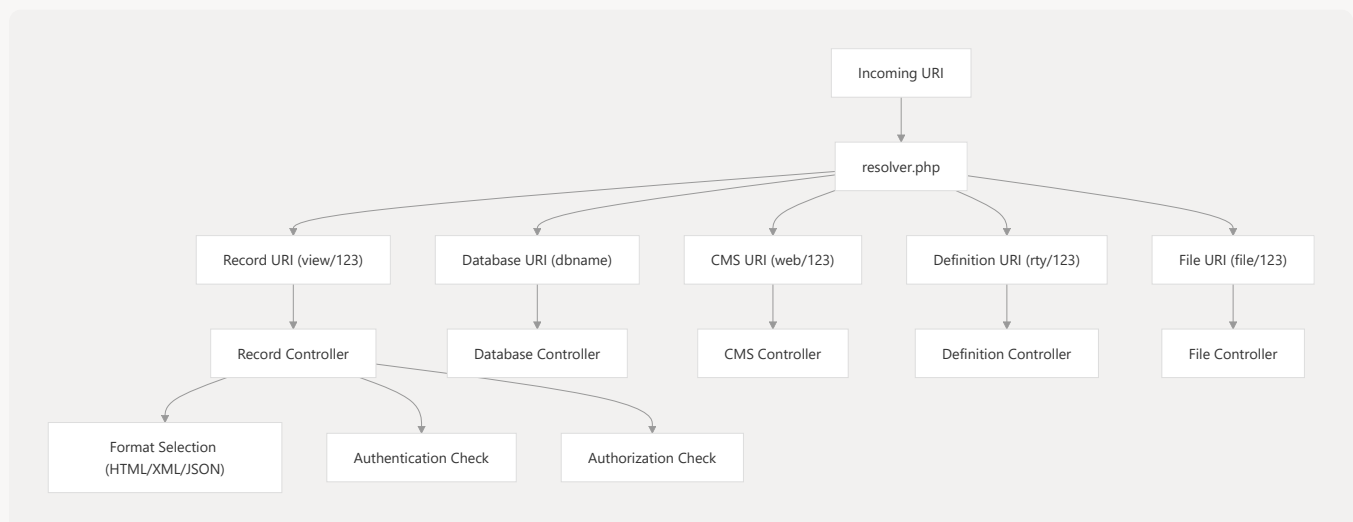
- XML/JSON
- HML
- CSV
- KML/GeoJSON (for geographic data)
- RDF
- Custom report formats

Sources: `redirects/resolver.php` 417-452 `admin/utilities/downloadInteractionLog.php` 38-217

URI Resolution System

Heurist implements a powerful URI resolution system to provide persistent identifiers for records and resources. This enables:

- Clean URLs for records: `domain.com/database/view/123`
- Cross-database record references: `domain.com/db/record/456-789` (where 456 is the database ID)
- CMS website access: `domain.com/database/web`
- Definition exports: `domain.com/db/rty/123` (for record types)



Sources: `redirects/resolver.php` 27-48 `redirects/resolver.php` 101-175

`redirects/resolver.php` 402-456

Conclusion

Heurist is a comprehensive knowledge management system with a modular architecture that supports flexible data structures, powerful search capabilities, and multiple ways to visualize and publish research data. Its client-server design with a central configuration system allows for easy